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Uplink multi-user multiple-input multiple-output (UL MU-MIMO) precoding using per-station feedback

Abstract

Various aspects relate generally to wireless communication and more particularly to beamforming techniques in wireless communication networks. Some aspects more specifically relate to beamforming techniques for uplink MU-MIMO communications, and the techniques can include UL MU-MIMO precoding using per-STA feedback. In some examples, an AP can support STA-side precoding for UL MU-MIMO communications by providing STAs with feedback representative of unitary matrices that correspond to block diagonal components of a matrix projection of channel matrices for the STAs in accordance with a linear equalizer. In some examples, the AP can trigger the STAs to transmit sounding packets concurrently, but can use the block diagonal matrix projection to process the various STAs' channels independently.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates generally to wireless communication, and more specifically, to beamforming techniques for wireless communications.

DESCRIPTION OF THE RELATED TECHNOLOGY

(2) A wireless local area network (WLAN) may be formed by one or more wireless access points (APs) that provide a shared wireless communication medium for use by multiple client devices also referred to as wireless stations (STAs). The basic building block of a WLAN conforming to the Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards is a Basic Service Set (BSS), which is managed by an AP. Each BSS is identified by a Basic Service Set Identifier (BSSID) that is advertised by the AP. An AP periodically broadcasts beacon frames to enable any STAs within wireless range of the AP to establish or maintain a communication link with the WLAN.

(3) In some WLANs, APs and STAs can transmit and receive wireless communications using multiple antennas. In order to increase the reliability and/or speed of wireless communications between them, APs and STAs can implement beamforming techniques. Such beamforming techniques can include techniques according to which STAs may apply precoding coefficients in order to beamform uplink multi-user multiple-input multiple-output (UL MU-MIMO) communications with APs.

SUMMARY

(4) The systems, methods and devices of this disclosure each have several innovative aspects, no single one of which is solely responsible for the desirable attributes disclosed herein.

(5) One innovative aspect of the subject matter described in this disclosure can be implemented in a wireless access point (AP). The wireless AP includes a processing system that includes processor circuitry and memory circuitry that stores code, the processing system configured to cause the wireless AP to transmit a first packet that indicates to each of a plurality of wireless stations (STAs) to transmit an uplink (UL) sounding packet, receive, in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively, and transmit a second packet that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA

(6) In some examples, the processing system can be further configured to cause the wireless AP to perform the measurements of the plurality of sounding packets over a plurality of subcarriers, obtain a respective channel matrix for each of the plurality of subcarriers in accordance with the measurements, obtain a respective projection of each channel matrix for each of the plurality of wireless STAs in accordance with a respective linear equalizer, and obtain the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs in accordance with respective block diagonal components of the respective projections. In some examples, the processing system can be further configured to cause the wireless AP to compress the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs using a respective Givens rotation operation to generate respective angles, wherein the per-STA UL beamforming feedback includes the respective angles.

(7) Another innovative aspect of the subject matter described in this disclosure can be implemented in a method for wireless communication by a wireless AP. The method includes transmitting a first packet that indicates to each of a plurality of wireless stations (STAs) to transmit an uplink (UL) sounding packet, receiving, in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively; and transmitting a second packet that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA.

(8) In some examples, the method can include performing the measurements of the plurality of sounding packets over a plurality of subcarriers, obtaining a respective channel matrix for each of the plurality of subcarriers in accordance with the measurements, obtaining a respective projection of each channel matrix for each of the plurality of wireless STAs in accordance with a respective linear equalizer, and obtaining the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs in accordance with respective block diagonal components of the respective projections. In some examples, the method can further comprise compressing the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs using a respective Givens rotation operation to generate respective angles, wherein the per-STA UL beamforming feedback includes the respective angles.

(9) Another innovative aspect of the subject matter described in this disclosure can be implemented in an apparatus. The apparatus includes means for transmitting a first packet that indicates to each of a plurality of wireless stations (STAs) to transmit an uplink (UL) sounding packet, means for receiving, in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively; and means for transmitting a second packet frame that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA, means for performing the measurements of the plurality of sounding packets over a plurality of subcarriers, means for obtaining a respective

channel matrix for each of the plurality of subcarriers in accordance with the measurements, means for obtaining a respective projection of each channel matrix for each of the plurality of wireless STAs in accordance with a respective linear equalizer; and means for obtaining the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs in accordance with respective block diagonal components of the respective projections.

(10) Another innovative aspect of the subject matter described in this disclosure can be implemented in non-transitory computer-readable medium storing instructions for communication by a wireless access point (AP). The instructions including code to transmit a first packet that indicates to each of a plurality of wireless stations (STAs) to transmit an uplink (UL) sounding packet, receive, in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively, and transmit a second packet that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA.

(11) Details of one or more implementations of the subject matter described in this disclosure are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages will become apparent from the description, the drawings and the claims. Note that the relative dimensions of the following figures may not be drawn to scale.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a pictorial diagram of an example wireless communication network.

(2) FIG. 2 shows an example protocol data unit (PDU) usable for communications between a wireless access point (AP) and one or more wireless stations (STAs).

(3) FIG. 3 shows an example physical layer (PHY) protocol data unit (PPDU) usable for communications between a wireless AP and one or more wireless STAs.

(4) FIG. 4 shows a hierarchical format of an example PPDU usable for communications between a wireless AP and one or more wireless STAs.

(5) FIG. 5 shows a block diagram illustrating an example beamforming feedback generation procedure.

(6) FIG. 6 shows a communication diagram illustrating an example communication exchange.

(7) FIG. 7 shows a block diagram illustrating an example operating environment.

(8) FIG. 8 shows a flowchart illustrating an example process performable by or at a wireless AP that supports UL MU-MIMO precoding using per-STA feedback.

(9) FIG. 9 shows a block diagram of an example wireless communication device that supports UL MU-MIMO precoding using per-STA feedback.

(10) Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

(11) The following description is directed to some particular examples for the purposes of describing innovative aspects of this disclosure. However, a person having ordinary skill in the art will readily recognize that the teachings herein can be applied in a multitude of different ways.

Some or all of the described examples may be implemented in any device, system or network that is capable of transmitting and receiving radio frequency (RF) signals according to one or more of the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standards, the IEEE 802.15 standards, the Bluetooth® standards as defined by the Bluetooth Special Interest Group (SIG), or the Long Term Evolution (LTE), 3G, 4G or 5G (New Radio (NR)) standards promulgated by the 3rd Generation Partnership Project (3GPP), among others. The described examples can be

implemented in any device, system or network that is capable of transmitting and receiving RF signals according to one or more of the following technologies or techniques: code division multiple access (CDMA), time division multiple access (TDMA), orthogonal frequency division multiplexing (OFDM), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), single-carrier FDMA (SC-FDMA), spatial division multiple access (SDMA), rate-splitting multiple access (RSMA), multi-user shared access (MUSA), single-user (SU) multiple-input multiple-output (MIMO) and multi-user (MU)-MIMO (MU-MIMO). The described examples also can be implemented using other wireless communication protocols or RF signals suitable for use in one or more of a wireless personal area network (WPAN), a wireless local area network (WLAN), a wireless wide area network (WWAN), a wireless metropolitan area network (WMAN), or an internet of things (IOT) network.

(12) Various aspects relate generally to wireless communication and more particularly to beamforming techniques in wireless communication networks. Some aspects more specifically relate to beamforming techniques for uplink (UL) MU-MIMO precoding in accordance with per-wireless station (STA) feedback representative of one or more respective unitary precoder matrices for each STA. In some examples, an access point (AP) can support STA-side precoding for UL MU-MIMO communications by providing each of a plurality of STAs with one or more respective unitary precoder matrices, or compressed angular representation of the one or more respective unitary precoder matrices, that corresponds to a subspace basis of a respective block diagonal component of modified channel matrices for subcarriers of a wireless channel. In which the modified channel matrix for each subcarrier projects a channel equalizer matrix for a channel associated with that subcarrier onto a estimated channel matrix associated with that subcarrier. In some examples, the channel equalizer matrix for a subcarrier may comprise a pseudo-inverse of the estimated channel matrix for the subcarrier. In some examples, the channel equalizer matrix for a subcarrier may comprise a regularized minimum mean square error (MMSE), or linear equalizer, of the estimated channel matrix for the subcarrier. In some examples, the AP can trigger the STAs to transmit UL sounding packets concurrently, but, by basing a number of long training fields (LTFs) in the trigger frame on a number of antennas received from each of the plurality of STAs (for example, in a capability exchange), can use null space or an equivalent to reduce inter-user and inter-stream interference, enabling individual streams to be processed and interference mitigated so that the AP can cause a plurality of STAs to transmit their sounding packets simultaneously. In some examples, the respective feedback representative of one or more respective unitary precoder matrices provided to each of the STAs may comprise compressed beamforming feedback that represents the one or more respective unitary precoder matrices. In some examples, the AP can use an existing format, such as a single-user compressed beamforming feedback (SU-CBF) format defined in IEEE 802.11ac, 802.11ax, or 802.11be, to provide each of a plurality of STAs with feedback representative of its one or more respective unitary precoder matrices. However, because feedback for the plurality of STAs may be determined jointly and therefore may not be modified in the same manner as SU-CBF, some examples may include an indication to each of the plurality of STAs (for example, in the trigger frame or a subsequent frame) of a modification restriction for using its one or more respective unitary precoder matrices. In some examples, the AP can use multi-user compressed beamforming (MU-CBF) In some examples, the compressed beamforming feedback for each STA can represent the application of compressed beamforming feedback matrix to the output of an SVD, Eigen Value Decomposition (EVD) or QR decomposition.

(13) Particular aspects of the subject matter described in this disclosure can be implemented to realize one or more of the following potential advantages. In some examples, the use of null space or an equivalent by an AP in the joint determination of feedback for a plurality of STAs in MU-MIMO precoding can reduce the computational costs to the AP associated with determining precoding coefficients. The disclosed techniques may also accommodate significant re-use of functional blocks of existing AP designs, such as singular value decomposition (SVD), compressed

V matrix (Cmp-V) formatting, and the like. By providing feedback to a plurality of STAs representative of one or more respective unitary precoder matrices for each STA, the AP can reduce overhead and leverage existing compression techniques and existing feedback formatting (such as SU-CBF), to provide efficient UL MU-MIMO beamforming feedback without imposing new requirements on STAs that already support UL MU-MIMO communications and beamforming.

(14) FIG. 1 shows a pictorial diagram of an example wireless communication network **100**.

According to some aspects, the wireless communication network **100** can be an example of a wireless local area network (WLAN) such as a Wi-Fi network. For example, the wireless communication network **100** can be a network implementing at least one of the IEEE 802.11 family of wireless communication protocol standards (such as defined by the IEEE 802.11-2020 specification or amendments thereof including, but not limited to, 802.11ay, 802.11ax, 802.11az, 802.11ba, 802.11bd, 802.11be, 802.11bf, and 802.11bn). In some other examples, the wireless communication network **100** can be an example of a cellular radio access network (RAN), such as a 5G or 6G RAN that implements one or more cellular protocols such as those specified in one or more 3GPP standards. In some other examples, the wireless communication network **100** can include a WLAN that functions in an interoperable or converged manner with one or more cellular RANs to provide greater or enhanced network coverage to wireless communication devices within the wireless communication network **100** or to enable such devices to connect to a cellular network's core, such as to access the network management capabilities and functionality offered by the cellular network core.

(15) The wireless communication network **100** may include numerous wireless communication devices including at least one wireless access point (AP) **102** and any number of wireless stations (STAs) **104**. While only one AP **102** is shown in FIG. 1, the wireless communication network **100** can include multiple APs **102**. The AP **102** can be or represent various different types of network entities including, but not limited to, a home networking AP, an enterprise-level AP, a single-frequency AP, a dual-band simultaneous (DBS) AP, a tri-band simultaneous (TBS) AP, a standalone AP, a non-standalone AP, a software-enabled AP (soft AP), and a multi-link AP (also referred to as an AP multi-link device (MLD)), as well as cellular (such as 3GPP, 4G LTE, 5G or 6G) base stations or other cellular network nodes such as a Node B, an evolved Node B (eNB), a gNB, a transmission reception point (TRP) or another type of device or equipment included in a radio access network (RAN), including Open-RAN (O-RAN) network entities, such as a central unit (CU), a distributed unit (DU) or a radio unit (RU).

(16) Each of the STAs **104** also may be referred to as a mobile station (MS), a mobile device, a mobile handset, a wireless handset, an access terminal (AT), a user equipment (UE), a subscriber station (SS), or a subscriber unit, among other examples. The STAs **104** may represent various devices such as mobile phones, other handheld or wearable communication devices, netbooks, notebook computers, tablet computers, laptops, Chromebooks, augmented reality (AR), virtual reality (VR), mixed reality (MR) or extended reality (XR) wireless headsets or other peripheral devices, wireless earbuds, other wearable devices, display devices (for example, TVs, computer monitors or video gaming consoles), video game controllers, navigation systems, music or other audio or stereo devices, remote control devices, printers, kitchen appliances (including smart refrigerators) or other household appliances, key fobs (for example, for passive keyless entry and start (PKES) systems), Internet of Things (IoT) devices, and vehicles, among other examples.

(17) A single AP **102** and an associated set of STAs **104** may be referred to as a basic service set (BSS), which is managed by the respective AP **102**. FIG. 1 additionally shows an example coverage area **108** of the AP **102**, which may represent a basic service area (BSA) of the wireless communication network **100**. The BSS may be identified by STAs **104** and other devices by a service set identifier (SSID), as well as a basic service set identifier (BSSID), which may be a medium access control (MAC) address of the AP **102**. The AP **102** may periodically broadcast beacon frames ("beacons") including the BSSID to enable any STAs **104** within wireless range of

the AP **102** to “associate” or re-associate with the AP **102** to establish a respective communication link **106** (hereinafter also referred to as a “Wi-Fi link”), or to maintain a communication link **106**, with the AP **102**. For example, the beacons can include an identification or indication of a primary channel used by the respective AP **102** as well as a timing synchronization function (TSF) for establishing or maintaining timing synchronization with the AP **102**. The AP **102** may provide access to external networks to various STAs **104** in the wireless communication network **100** via respective communication links **106**.

(18) To establish a communication link **106** with an AP **102**, each of the STAs **104** is configured to perform passive or active scanning operations (“scans”) on frequency channels in one or more frequency bands (for example, the 2.4 GHz, 5 GHz, 6 GHz, 45 GHz, or 60 GHz bands). To perform passive scanning, a STA **104** listens for beacons, which are transmitted by respective APs **102** at periodic time intervals referred to as target beacon transmission times (TBTTs). To perform active scanning, a STA **104** generates and sequentially transmits probe requests on each channel to be scanned and listens for probe responses from APs **102**. Each STA **104** may identify, determine, ascertain, or select an AP **102** with which to associate in accordance with the scanning information obtained through the passive or active scans, and to perform authentication and association operations to establish a communication link **106** with the selected AP **102**. The selected AP **102** assigns an association identifier (AID) to the STA **104** at the culmination of the association operations, which the AP **102** uses to track the STA **104**.

(19) As a result of the increasing ubiquity of wireless networks, a STA **104** may have the opportunity to select one of many BSSs within range of the STA **104** or to select among multiple APs **102** that together form an extended service set (ESS) including multiple connected BSSs. For example, the wireless communication network **100** may be connected to a wired or wireless distribution system that may enable multiple APs **102** to be connected in such an ESS. As such, a STA **104** can be covered by more than one AP **102** and can associate with different APs **102** at different times for different transmissions. Additionally, after association with an AP **102**, a STA **104** also may periodically scan its surroundings to find a more suitable AP **102** with which to associate. For example, a STA **104** that is moving relative to its associated AP **102** may perform a “roaming” scan to find another AP **102** having more desirable network characteristics such as a greater received signal strength indicator (RSSI) or a reduced traffic load.

(20) In some cases, STAs **104** may form networks without APs **102** or other equipment other than the STAs **104** themselves. One example of such a network is an ad hoc network (or wireless ad hoc network). Ad hoc networks may alternatively be referred to as mesh networks or peer-to-peer (P2P) networks. In some cases, ad hoc networks may be implemented within a larger network such as the wireless communication network **100**. In such examples, while the STAs **104** may be capable of communicating with each other through the AP **102** using communication links **106**, STAs **104** also can communicate directly with each other via direct wireless communication links **110**.

Additionally, two STAs **104** may communicate via a direct communication link **110** regardless of whether both STAs **104** are associated with and served by the same AP **102**. In such an ad hoc system, one or more of the STAs **104** may assume the role filled by the AP **102** in a BSS. Such a STA **104** may be referred to as a group owner (GO) and may coordinate transmissions within the ad hoc network. Examples of direct wireless communication links **110** include Wi-Fi Direct connections, connections established by using a Wi-Fi Tunneled Direct Link Setup (TDLS) link, and other P2P group connections.

(21) In some networks, the AP **102** or the STAs **104**, or both, may support applications associated with high throughput or low-latency requirements, or may provide lossless audio to one or more other devices. For example, the AP **102** or the STAs **104** may support applications and use cases associated with ultra-low-latency (ULL), such as ULL gaming, or streaming lossless audio and video to one or more personal audio devices (such as peripheral devices) or AR/VR/MR/XR headset devices. In scenarios in which a user uses two or more peripheral devices, the AP **102** or

the STAs **104** may support an extended personal audio network enabling communication with the two or more peripheral devices. Additionally, the AP **102** and STAs **104** may support additional ULL applications such as cloud-based applications (such as VR cloud gaming) that have ULL and high throughput requirements.

(22) As indicated above, in some implementations, the AP **102** and the STAs **104** may function and communicate (via the respective communication links **106**) according to one or more of the IEEE 802.11 family of wireless communication protocol standards. These standards define the WLAN radio and baseband protocols for the physical (PHY) and MAC layers. The AP **102** and STAs **104** transmit and receive wireless communications (hereinafter also referred to as “Wi-Fi communications” or “wireless packets”) to and from one another in the form of PHY protocol data units (PPDUs).

(23) Each PPDU is a composite structure that includes a PHY preamble and a payload that is in the form of a PHY service data unit (PSDU). The information provided in the preamble may be used by a receiving device to decode the subsequent data in the PSDU. In instances in which a PPDU is transmitted over a bonded or wideband channel, the preamble fields may be duplicated and transmitted in each of multiple component channels. The PHY preamble may include both a legacy portion (or “legacy preamble”) and a non-legacy portion (or “non-legacy preamble”). The legacy preamble may be used for packet detection, automatic gain control and channel estimation, among other uses. The legacy preamble also may generally be used to maintain compatibility with legacy devices. The format of, coding of, and information provided in the non-legacy portion of the preamble is associated with the particular IEEE 802.11 wireless communication protocol to be used to transmit the payload.

(24) The APs **102** and STAs **104** in the WLAN **100** may transmit PPDUs over an unlicensed spectrum, which may be a portion of spectrum that includes frequency bands traditionally used by Wi-Fi technology, such as the 2.4 GHz, 5 GHz, 6 GHz, 45 GHz, and 60 GHz bands. Some examples of the APs **102** and STAs **104** described herein also may communicate in other frequency bands that may support licensed or unlicensed communications. For example, the APs **102** or STAs **104**, or both, also may be capable of communicating over licensed operating bands, where multiple operators may have respective licenses to operate in the same or overlapping frequency ranges. Such licensed operating bands may map to or be associated with frequency range designations of FR1 (410 MHz-7.125 GHz), FR2 (24.25 GHz-52.6 GHz), FR3 (7.125 GHz-24.25 GHz), FR4a or FR4-1 (52.6 GHz-71 GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz).

(25) Each of the frequency bands may include multiple sub-bands and frequency channels (also referred to as subchannels). For example, PPDUs conforming to the IEEE 802.11n, 802.11ac, 802.11ax, 802.11be and 802.11bn standard amendments may be transmitted over one or more of the 2.4 GHz, 5 GHz, or 6 GHz bands, each of which is divided into multiple 20 MHz channels. As such, these PPDUs are transmitted over a physical channel having a minimum bandwidth of 20 MHz, but larger channels can be formed through channel bonding. For example, PPDUs may be transmitted over physical channels having bandwidths of 40 MHz, 80 MHz, 160 MHz, 240 MHz, 320 MHz, 480 MHz, or 640 MHz by bonding together multiple 20 MHz channels.

(26) FIG. 2 shows an example protocol data unit (PDU) **200** usable for wireless communication between a wireless AP and one or more wireless STAs. For example, the AP and STAs may be examples of the AP **102** and the STAs **104** described with reference to FIG. 1. The PDU **200** can be configured as a PPDU. As shown, the PDU **200** includes a PHY preamble **202** and a PHY payload **204**. For example, the preamble **202** may include a legacy portion that itself includes a legacy short training field (L-STF) **206**, which may consist of two symbols, a legacy long training field (L-LTF) **208**, which may consist of two symbols, and a legacy signal field (L-SIG) **210**, which may consist of two symbols. The legacy portion of the preamble **202** may be configured according to the IEEE 802.11a wireless communication protocol standard. The preamble **202** also may include a non-legacy portion including one or more non-legacy fields **212**, for example, conforming to one or

more of the IEEE 802.11 family of wireless communication protocol standards.

(27) The L-STF **206** generally enables a receiving device (such as AP **102** or STA **104**) to perform coarse timing and frequency tracking and automatic gain control (AGC). The L-LTF **208** generally enables the receiving device to perform fine timing and frequency tracking and also to perform an initial estimate of the wireless channel. The L-SIG **210** generally enables the receiving device to determine (for example, obtain, select, identify, detect, ascertain, calculate, or compute) a duration of the PDU and to use the determined duration to avoid transmitting on top of the PDU. The legacy portion of the preamble, including the L-STF **206**, the L-LTF **208** and the L-SIG **210**, may be modulated according to a binary phase shift keying (BPSK) modulation scheme. The payload **204** may be modulated according to a BPSK modulation scheme, a quadrature BPSK (Q-BPSK) modulation scheme, a quadrature amplitude modulation (QAM) modulation scheme, or another appropriate modulation scheme. The payload **204** may include a PSDU including a data field (DATA) **214** that, in turn, may carry higher layer data, for example, in the form of MAC protocol data units (MPDUs) or an aggregated MPDU (A-MPDU).

(28) FIG. 3 shows an example physical layer (PHY) protocol data unit (PPDU) **350** usable for communications between a wireless AP and one or more wireless STAs. For example, the AP and STAs may be examples of the AP **102** and the STAs **104** described with reference to FIG. 1. As shown, the PPDU **350** includes a PHY preamble, that includes a legacy portion **352** and a non-legacy portion **354**, and a payload **356** that includes a data field **374**. The legacy portion **352** of the preamble includes an L-STF **358**, an L-LTF **360**, and an L-SIG **362**. The non-legacy portion **354** of the preamble includes a repetition of L-SIG (RL-SIG) **364** and multiple wireless communication protocol version-dependent signal fields after RL-SIG **364**. For example, the non-legacy portion **354** may include a universal signal field **366** (referred to herein as “U-SIG **366**”) and an EHT signal field **368** (referred to herein as “EHT-SIG **368**”). The presence of RL-SIG **364** and U-SIG **366** may indicate to EHT- or later version-compliant STAs **104** that the PPDU **350** is an EHT PPDU or a PPDU conforming to any later (post-EHT) version of a new wireless communication protocol conforming to a future IEEE 802.11 wireless communication protocol standard. One or both of U-SIG **366** and EHT-SIG **368** may be structured as, and carry version-dependent information for, other wireless communication protocol versions associated with amendments to the IEEE family of standards beyond EHT. For example, U-SIG **366** may be used by a receiving device (such as the AP **102** or the STA **104**) to interpret bits in one or more of EHT-SIG **368** or the data field **374**. Like L-STF **358**, L-LTF **360**, and L-SIG **362**, the information in U-SIG **366** and EHT-SIG **368** may be duplicated and transmitted in each of the component 20 MHz channels in instances involving the use of a bonded channel.

(29) The non-legacy portion **354** further includes an additional short training field **370** (referred to herein as “EHT-STF **370**,” although it may be structured as, and carry version-dependent information for, other wireless communication protocol versions beyond EHT) and one or more additional long training fields **372** (referred to herein as “EHT-LTFs **372**,” although they may be structured as, and carry version-dependent information for, other wireless communication protocol versions beyond EHT). EHT-STF **370** may be used for timing and frequency tracking and AGC, and EHT-LTF **372** may be used for more refined channel estimation.

(30) EHT-SIG **368** may be used by an AP **102** to identify and inform one or multiple STAs **104** that the AP **102** has scheduled uplink (UL) or downlink (DL) resources for them. EHT-SIG **368** may be decoded by each compatible STA **104** served by the AP **102**. EHT-SIG **368** may generally be used by the receiving device to interpret bits in the data field **374**. For example, EHT-SIG **368** may include resource unit (RU) allocation information, spatial stream configuration information, and per-user (for example, STA-specific) signaling information. Each EHT-SIG **368** may include a common field and at least one user-specific field. In the context of OFDMA, the common field can indicate RU distributions to multiple STAs **104**, indicate the RU assignments in the frequency domain, indicate which RUs are allocated for MU-MIMO transmissions and which RUs

correspond to OFDMA transmissions, and the number of users in allocations, among other examples. The user-specific fields are assigned to particular STAs **104** and carry STA-specific scheduling information such as user-specific MCS values and user-specific RU allocation information. Such information enables the respective STAs **104** to identify and decode corresponding RUs in the associated data field **374**.

(31) FIG. **4** shows a hierarchical format of an example PPDU usable for communications between a wireless AP and one or more wireless STAs. For example, the AP and STAs may be examples of the AP **102** and the STAs **104** described with reference to FIG. **1**. As described, each PPDU **400** includes a PHY preamble **402** and a PSDU **404**. Each PSDU **404** may represent (or “carry”) one or more MAC protocol data units (MPDUs) **416**. For example, each PSDU **404** may carry an aggregated MPDU (A-MPDU) **406** that includes an aggregation of multiple A-MPDU subframes **408**. Each A-MPDU subframe **406** may include an MPDU frame **410** that includes a MAC delimiter **412** and a MAC header **414** prior to the accompanying MPDU **416**, which includes the data portion (“payload” or “frame body”) of the MPDU frame **410**. Each MPDU frame **410** also may include a frame check sequence (FCS) field **418** for error detection (for example, the FCS field may include a cyclic redundancy check (CRC)) and padding bits **420**. The MPDU **416** may carry one or more MAC service data units (MSDUs) **416**. For example, the MPDU **416** may carry an aggregated MSDU (A-MSDU) **422** including multiple A-MSDU subframes **424**. Each A-MSDU subframe **424** contains a corresponding MSDU **430** preceded by a subframe header **428** and in some cases followed by padding bits **432**.

(32) Referring back to the MPDU frame **410**, the MAC delimiter **412** may serve as a marker of the start of the associated MPDU **416** and indicate the length of the associated MPDU **416**. The MAC header **414** may include multiple fields containing information that defines or indicates characteristics or attributes of data encapsulated within the frame body **416**. The MAC header **414** includes a duration field indicating a duration extending from the end of the PPDU until at least the end of an acknowledgment (ACK) or Block ACK (BA) of the PPDU that is to be transmitted by the receiving wireless communication device. The use of the duration field serves to reserve the wireless medium for the indicated duration, and enables the receiving device to establish its network allocation vector (NAV). The MAC header **414** also includes one or more fields indicating addresses for the data encapsulated within the frame body **416**. For example, the MAC header **414** may include a combination of a source address, a transmitter address, a receiver address or a destination address. The MAC header **414** may further include a frame control field containing control information. The frame control field may specify a frame type, for example, a data frame, a control frame, or a management frame.

(33) APs and STAs (for example, the AP **102** and the STAs **104** described with reference to FIG. **1**) that include multiple antennas may support various diversity schemes. For example, spatial diversity may be used by one or both of a transmitting device (such as either AP **102** or STA **104**) or a receiving device (such as either AP **102** or STA **104**) to increase the robustness of a transmission. For example, to implement a transmit diversity scheme, a transmitting device may transmit the same data redundantly over two or more antennas.

(34) APs **102** and STAs **104** that include multiple antennas also may support space-time block coding (STBC). With STBC, a transmitting device also transmits multiple copies of a data stream across multiple antennas to exploit the various received versions of the data to increase the likelihood of decoding the correct data. More specifically, the data stream to be transmitted is encoded in blocks, which are distributed among the spaced antennas and across time. Generally, STBC can be used when the number $N_{\text{sub.Tx}}$ of transmit antennas exceeds the number $N_{\text{sub.SS}}$ of spatial streams. The $N_{\text{sub.SS}}$ spatial streams may be mapped to a number $N_{\text{sub.STS}}$ of space-time streams, which are then mapped to $N_{\text{sub.Tx}}$ transmit chains.

(35) APs **102** and STAs **104** that include multiple antennas also may support spatial multiplexing, which may be used to increase the spectral efficiency and the resultant throughput of a

transmission. To implement spatial multiplexing, the transmitting device divides the data stream into a number $N_{\text{sub.SS}}$ of separate, independent spatial streams. The spatial streams are then separately encoded and transmitted in parallel via the multiple $N_{\text{sub.Tx}}$ transmit antennas.

(36) APs **102** and STAs **104** that include multiple antennas also may support beamforming. Beamforming generally refers to the steering of the energy of a transmission in the direction of a target receiver. Beamforming may be used both in a single-user (SU) context, for example, to improve a signal-to-noise ratio (SNR), as well as in a multi-user (MU) context, for example, to enable MU-MIMO transmissions (also referred to as spatial division multiple access (SDMA)). In the MU-MIMO context, beamforming may additionally or alternatively involve the nulling out of energy in the directions of other receiving devices. To perform SU beamforming or MU-MIMO, a transmitting device, referred to as the beamformer, transmits a signal from each of multiple antennas. The beamformer configures the amplitudes and phase shifts between the signals transmitted from the different antennas such that the signals add constructively along particular directions towards the intended receiver (referred to as the beamformee) or add destructively in other directions towards other devices to mitigate interference in a MU-MIMO context. The manner in which the beamformer configures the amplitudes and phase shifts depends on channel state information (CSI) associated with the wireless channels over which the beamformer intends to communicate with the beamformee.

(37) To obtain the CSI necessary for beamforming, the beamformer may perform a channel sounding procedure with the beamformee. For example, the beamformer may transmit one or more sounding signals (for example, in the form of a null data packet (NDP)) to the beamformee. An NDP is a PPDU without any data field. The beamformee may then perform measurements for each of the $N_{\text{sub.Tx}} \times N_{\text{sub.Rx}}$ sub-channels corresponding to all of the transmit antenna and receive antenna pairs associated with the sounding signal. The beamformee generates a feedback matrix associated with the channel measurements and, typically, compresses the feedback matrix before transmitting the feedback to the beamformer. The beamformer may then generate a precoding (or “steering”) matrix for the beamformee associated with the feedback and use the steering matrix to precode the data streams to configure the amplitudes and phase shifts for subsequent transmissions to the beamformee. The beamformer may use the steering matrix to determine (for example, identify, detect, ascertain, calculate, or compute) how to transmit a signal on each of its antennas to perform beamforming. For example, the steering matrix may be indicative of a phase shift, power level, etc. to use to transmit a respective signal on each of the beamformer's antennas.

(38) When performing beamforming, the transmitting beamforming array gain is logarithmically proportional to the ratio of $N_{\text{sub.Tx}}$ to $N_{\text{sub.SS}}$. As such, it is generally desirable, within other constraints, to increase the number $N_{\text{sub.Tx}}$ of transmit antennas when performing beamforming to increase the gain. It is also possible to more accurately direct transmissions or nulls by increasing the number of transmit antennas. This is especially advantageous in MU transmission contexts in which it is particularly important to reduce inter-user interference.

(39) To increase an AP **102**'s spatial multiplexing capability, an AP **102** may need to support an increased number of spatial streams (such as up to 16 spatial streams). However, supporting additional spatial streams may result in increased CSI feedback overhead. Implicit CSI acquisition techniques may avoid CSI feedback overhead by taking advantage of the assumption that the UL and DL channels have reciprocal impulse responses (that is, that there is channel reciprocity). For example, the CSI feedback overhead may be reduced using an implicit channel sounding procedure such as an implicit beamforming report (BFR) technique (such as where STAs **104** transmit NDP sounding packets in the UL while the AP **102** measures the channel) because no BFRs are sent. Once the AP **102** receives the NDPs, it may implicitly assess the channels for each of the STAs **104** and use the channel assessments to configure steering matrices. In order to mitigate hardware mismatches that could break the channel reciprocity on the UL and DL (such as the baseband-to-RF and RF-to-baseband chains not being reciprocal), the AP **102** may implement a calibration

method to compensate for the mismatch between the UL and the DL channels. For example, the AP **102** may select a reference antenna, transmit a pilot signal from each of its antennas, and estimate baseband-to-RF gain for each of the non-reference antennas relative to the reference antenna.

(40) In some examples, multiple APs **102** may simultaneously transmit signaling or communications to a single STA **104** utilizing a distributed MU-MIMO scheme. Examples of such a distributed MU-MIMO transmission include coordinated beamforming (CBF) and joint transmission (JT). With CBF, signals (such as data streams) for a given STA **104** may be transmitted by only a single AP **102**. However, the coverage areas of neighboring APs may overlap, and signals transmitted by a given AP **102** may reach the STAs in OBSSs associated with neighboring APs as OBSS signals. CBF allows multiple neighboring APs to transmit simultaneously while minimizing or avoiding interference, which may result in more opportunities for spatial reuse. More specifically, using CBF techniques, an AP **102** may beamform signals to in-BSS STAs **104** while forming nulls in the directions of STAs in OBSSs such that any signals received at an OBSS STA are of sufficiently low power to limit the interference at the STA. To accomplish this, an inter-BSS coordination set may be defined between the neighboring APs, which contains identifiers of all APs and STAs participating in CBF transmissions.

(41) With JT, signals for a given STA **104** may be transmitted by multiple coordinated APs **102**. For the multiple APs **102** to concurrently transmit data to a STA **104**, the multiple APs **102** may all need a copy of the data to be transmitted to the STA **104**. Accordingly, the APs **102** may need to exchange the data among each other for transmission to a STA **104**. With JT, the combination of antennas of the multiple APs **102** transmitting to one or more STAs **104** may be considered as one large antenna array (which may be represented as a virtual antenna array) used for beamforming and transmitting signals. In combination with MU-MIMO techniques, the multiple antennas of the multiple APs **102** may be able to transmit data via multiple spatial streams. Accordingly, each STA **104** may receive data via one or more of the multiple spatial streams.

(42) In some implementations, the AP **102** and STAs **104** can support various multi-user communications; that is, concurrent transmissions from one device to each of multiple devices (for example, multiple simultaneous downlink communications from an AP **102** to corresponding STAs **104**), or concurrent transmissions from multiple devices to a single device (for example, multiple simultaneous uplink transmissions from corresponding STAs **104** to an AP **102**). As an example, in addition to MU-MIMO, the AP **102** and STAs **104** may support OFDMA. OFDMA is in some aspects a multi-user version of OFDM.

(43) In OFDMA schemes, the available frequency spectrum of the wireless channel may be divided into multiple resource units (RUs) each including multiple frequency subcarriers (also referred to as “tones”). Different RUs may be allocated or assigned by an AP **102** to different STAs **104** at particular times. The sizes and distributions of the RUs may be referred to as an RU allocation. In some examples, RUs may be allocated in 2 MHz intervals, and as such, the smallest RU may include 26 tones consisting of 24 data tones and 2 pilot tones. Consequently, in a 20 MHz channel, up to 9 RUs (such as 2 MHz, 26-tone RUs) may be allocated (because some tones are reserved for other purposes). Similarly, in a 160 MHz channel, up to 74 RUs may be allocated. Other tone RUs also may be allocated, such as 52 tone, 106 tone, 242 tone, 484 tone and 996 tone RUs. Adjacent RUs may be separated by a null subcarrier (such as a DC subcarrier), for example, to reduce interference between adjacent RUs, to reduce receiver DC offset, and to avoid transmit center frequency leakage.

(44) For UL MU transmissions, an AP **102** can transmit a trigger frame to initiate and synchronize an UL OFDMA or UL MU-MIMO transmission from multiple STAs **104** to the AP **102**. Such trigger frames may thus enable multiple STAs **104** to send UL traffic to the AP **102** concurrently in time. A trigger frame may address one or more STAs **104** through respective association identifiers (AIDs), and may assign each AID (and thus each STA **104**) one or more RUs that can be used to send UL traffic to the AP **102**. The AP also may designate one or more random access (RA) RUs

that unscheduled STAs **104** may contend for.

(45) FIG. 5 shows a block diagram illustrating an example beamforming feedback generation procedure **500**. According to the beamforming feedback generation procedure **500**, compressed beamforming feedback (CBF) tones for a UL MU-MIMO channel can be selected at **502**.

Following MMSE processing at **504**, a block diagonal (BD) matrix projection can be performed at **506** for a plurality of users (STAs) 1 to M. Here, the BD matrix projection may act as null space projection or equivalent to help reduce inter-user and inter-stream interference. For each of the users (STAs) 1 to M, singular value decomposition (SVD) can be performed at **508** for the plurality of subcarriers (tones) selected in block **502** to obtain a respective compressed V matrix (Cmp-V) at **510**. Based on the component vectors at **510**, a compressed beamforming feedback report can be generated at **512**. It can be noted that, in some embodiments, the CBF report generated in **512** may contain a separate beamforming feedback report for users (STAs) 1 to M, which may be encapsulated together to be sent together in the subsequent frame to the users (STAs) or may be sent individually over subsequent frames. It can also be noted that in one embodiment SVD may be substituted by other channel decomposition algorithms such as EVD (Eigen Value Decomposition) or QR decomposition.

(46) FIG. 6 shows a communication diagram illustrating an example communication exchange **600**. According to communication exchange **600**, a wireless AP can transmit a trigger frame for UL sounding. In response to the trigger frame, a plurality of wireless STAs 1 to M can concurrently transmit respective UL sounding packets—which can be null data packets (NDPs)—to the wireless AP. The wireless AP can then transmit per-user compressed beamforming feedback (CBF) to the wireless STAs. In some examples, the wireless AP can transmit the per-user CBF in a downlink (DL) orthogonal frequency division multiple access (OFDMA) mode. In some examples, the per-user CBF can include an acknowledgment trigger, and the wireless STAs can transmit respective acknowledgments to the wireless AP to acknowledge receipt of the per-user CBF. The wireless AP can subsequently transmit a trigger frame to cause the wireless STAs to perform UL beamformed data transmission according to beamforming feedback provided to them in the per-user CBF.

(47) FIG. 7 shows a block diagram illustrating an example operating environment **700**. In the operating environment **700**, a wireless AP **702** can implement UL MU-MIMO precoding using per-STA feedback in supporting of beamforming on the part of one or more wireless STAs **704**. According to aspects of the disclosure, in the operating environment **700**, the wireless AP **702** can implement the beamforming feedback generation procedure **500** of FIG. 5. According to aspects of the disclosure, in the operating environment **700**, the wireless AP **702** can communicate with the wireless STAs **704** in accordance with the communication exchange **600** of FIG. 6.

(48) In the operating environment **700**, the wireless AP **702** can support precoding according to a UL MU-MIMO minimum mean square error (MMSE) precoding model. According to the UL MU-MIMO MMSE precoding model, for a subcarrier k, for a total of M users, an estimated channel can be represented by an estimated channel matrix $H_{\text{sup.k}}$ according to Equation (1) as follows:

$$(49) \quad H^k = [H_1^k \dots H_m^k] \in N_r \times \text{Math}_{m=1}^N N_{t,m} \quad (1)$$

where $N_{\text{sub.t,m}}$ represents the number of transmit antennas of the $m_{\text{sup.th}}$ user.

(50) The pseudo-inverse of the estimated channel matrix $H_{\text{sup.k}}$ can be represented by $G_{\text{sup.k}}$ according to Equation (2) as follows:

$$(51) \quad G^k = H^{k,*} (H^k H^{k,*})^{-1} = \begin{bmatrix} G_1^k \\ \text{Math.} \\ G_M^k \end{bmatrix} \in \text{Math}_{m=1}^M N_{t,m} \times N_r \quad (2)$$

(52) $G_{\text{sup.k}} H_{\text{sup.k}}$ (which may correspond to the plurality of modified channel matrices described elsewhere herein) can then be written explicitly according to Equation (3) as follows:

$$(53) \quad G^k H^k = \begin{bmatrix} G_1^k & \text{.Math.} & G_1^k H_1^k & \text{.Math.} & G_1^k H_M^k \\ \text{.Math.} & [H_1^k \text{.Math.} H_m^k] & \text{.Math.} & \ddots & \text{.Math.} \\ G_M^k & & G_M^k H_1^k & \text{.Math.} & G_M^k H_M^k \end{bmatrix} \quad (3)$$

(54) For each of users $m=1$ to M , wireless AP **702** can find the respective subspace basis of $\check{G}_{\text{sub.m.sup.kH.sub.m.sup.k}}$ and feedback the corresponding unitary matrix or compressed angular representation of the unitary matrix to the wireless STA **704** corresponding to the user m .

(55) According to aspects of the disclosure, this process may correspond to the beamforming feedback generation procedure **500** of FIG. 5, described above. In some examples, the determination of the pseudo-inverse of the estimated channel matrix using Equation (2) may correspond to MMSE **504**. In some examples, this estimated channel matrix pseudo-inverse may be replaced with the regularized MMSE inverse, or linear equalizer. In some examples, the determination of $G_{\text{sup.kH.sup.k}}$ may correspond to the BD Matrix Projection **506** in which $G_{\text{sup.kH.sup.k}}$ the channel equalizer matrix for a channel associated with a subcarrier k ($G_{\text{sup.k}}$) is multiplied with a channel matrix associated with that subcarrier ($H_{\text{sup.k}}$). Put differently, $G_{\text{sup.kH.sup.k}}$ may represent a respective projection of each channel matrix for each of the plurality of wireless STAs ($H_{\text{sup.k}}$) in accordance with a respective linear equalizer ($G_{\text{sup.k}}$). The unitary matrix for a given user, m , may be obtained by, for example, applying SVD to the corresponding block diagonal component of $G_{\text{sup.kH.sup.k}}$, $\check{G}_{\text{sub.m.sup.kH.sub.m.sup.k}}$, as indicated at **508** of FIG. 5. In some examples, the unitary matrix may be compressed. In some examples, this compression may utilize cmpV format, as indicated at **510** of FIG. 5.

(56) According to aspects of the disclosure, the wireless AP **702** can transmit a frame **706** on a wireless channel between the wireless AP **702** and the wireless STAs **704** in conjunction with UL MU-MIMO beamforming in the operating environment **700**. In some examples, the frame **706** can be a trigger frame. According to aspects of the disclosure, the frame **706** can identify the wireless STAs **704** as a plurality of wireless STAs that are to perform UL sounding of the wireless channel.

(57) According to aspects of the disclosure, the wireless AP **702** can receive via the wireless channel, associated with the transmission of the frame **706**, multiple sounding packets **708**. In some examples, the sounding packets **708** can include a respective sounding packet **708** from each of the wireless STAs **704**.

(58) According to aspects of the disclosure, the wireless AP **702** can transmit a frame **710** on the wireless channel associated with the receipt of the sounding packets **708**. In some examples, the frame **710** can include per-STA UL beamforming feedback **712** for a wireless STA **704** among the wireless STAs **704**. In some examples, the frame **710** can include a compressed beamforming report field that includes the per-STA UL beamforming feedback **712**. In some examples, the frame **710** can be a compressed beamforming/channel quality indication (CQI) frame. In some examples, the frame **710** can include an acknowledgment trigger. In some examples, the per-STA UL beamforming feedback **712** for the wireless STA **704** can include a plurality of compressed beamforming matrices, where each of the plurality of compressed beamforming feedback matrices is associated with a respective one of a plurality of subcarriers of the wireless channel. In some such examples, the compressed beamforming feedback matrices can be ordered according to the frequencies of their associated subcarriers, from lowest frequency to highest frequency. In some examples, each of the plurality of compressed beamforming feedback matrices can include a respective angle sequence (for example, for a Givens rotation). In some examples, each angle sequence can be a sequence of angles including ϕ angles and ψ angles.

(59) According to aspects of the disclosure, the wireless AP **702** can obtain the plurality of compressed beamforming feedback matrices with reference to block diagonal components of a plurality of modified channel matrices including a respective associated modified channel matrix for each of the plurality of subcarriers. In some examples, the wireless AP **702** can obtain the

plurality of compressed beamforming feedback matrices from the block diagonal components of the plurality of modified channel matrices using singular value decomposition (SVD) or QR decomposition. In some examples, a respective associated modified channel matrix for at least one of the plurality of subcarriers can include a projection of a channel equalizer matrix, or linear equalizer, for a channel associated with that subcarrier and a channel matrix associated with that subcarrier.

(60) In some examples, the wireless AP **702** can transmit a frame **714** on the wireless channel to trigger a wireless station **704** for which it provided the per-STA UL beamforming feedback **712** to perform a beamformed UL data transmission according to the per-STA UL beamforming feedback **712**.

(61) FIG. **8** shows a flowchart illustrating an example process **800** performable by or at a wireless AP that supports UL MU-MIMO precoding using per-STA feedback. The operations of the process **800** may be implemented by a wireless AP or its components as described herein. For example, the process **800** may be performed by a wireless communication device, such as the wireless communication device **900** described with reference to FIG. **9**, operating as or within a wireless AP. In some examples, the process **800** may be performed by a wireless AP such as one of the APs **102** described with reference to FIG. **1**.

(62) In some examples, in block **802**, the wireless AP can transmit a first packet that indicates to each of a plurality of wireless STAs to transmit an UL sounding packet. For example, in the operating environment **700** of FIG. **7**, the wireless AP **702** can transmit the frame **706** on the wireless channel between the wireless AP **702** and the wireless STAs **704**, and the frame **706** can identify the wireless STAs **704** as a plurality of wireless STAs that are to perform UL sounding of the wireless channel. In some examples, the first packet can include a trigger configured to cause the plurality of wireless STAs to transmit the plurality of sounding packets. As such, in some examples, the first frame (for example, frame **706**) can be a trigger frame.

(63) In some examples, in block **804**, the wireless AP can receive in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively. For example, in the operating environment **700** of FIG. **7**, the wireless AP **702** can receive, via the wireless channel between the wireless AP **702** and the wireless STAs **704**, the sounding packets **708**, and the sounding packets **708** can include a respective sounding packet **708** from each of the wireless STAs **704**.

(64) In some examples, in block **806**, the wireless AP can transmit a second packet that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA. For example, in the operating environment **700** of FIG. **7**, the wireless AP **702** can transmit the frame **710** on the wireless channel between the wireless AP **702** and the wireless STAs **704** associated with the receipt of the sounding packets **708**, and the frame **710** can comprise a packet that includes per-STA UL beamforming feedback **712** for a wireless STA **704** among the wireless STAs **704**.

(65) In some examples, the wireless AP can perform the measurements of the plurality of sounding packets over a plurality of subcarriers; obtain a respective channel matrix for each of the plurality of subcarriers in accordance with the measurements; obtain a respective projection of each channel matrix for each of the plurality of wireless STAs in accordance with a respective linear equalizer; and obtain the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs in accordance with respective block diagonal components of the respective projections. For example, in the operating environment **700** of FIG. **7**, the wireless AP **702** can perform measurements of the plurality of sounding packets **708** over a plurality of subcarriers, k , and for a total of M STAs **704**. The wireless AP **702** can then obtain a respective channel matrix, $H_{\text{sup},k}$, for each of the plurality of subcarriers in accordance with the

measurements, and obtain a respective projection of each channel matrix, $G_{sup.kH.sup.k}$, for each of the plurality of wireless STAs in accordance with a respective linear equalizer, $G_{sup.k}$, in the manner described above with respect to BD Matrix Projection **506** of FIG. 5, for example. The wireless AP **702** can then obtain the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs in accordance with respective block diagonal components of the respective projections.

(66) In some examples, the wireless AP can compress the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs using a respective Givens rotation operation to generate respective angles, wherein the per-STA UL beamforming feedback includes the respective angles. In some examples, the wireless AP can obtain the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs, the processing system is configured to cause the wireless AP to perform a singular value decomposition (SVD), Eigen Value Decomposition (EVD) or QR decomposition operation on the respective block diagonal components of the respective projections. In some examples, the respective projection of each channel matrix for each of the plurality of wireless STAs comprises a null space matrix projection. In some examples, the null space matrix projection comprises a null space block diagonal matrix projection.

(67) In some examples, the second packet includes a respective compressed beamforming report field for each of the plurality of wireless STAs that includes the per-STA UL beamforming feedback for the plurality of wireless stations. In some examples, the second packet further includes a respective per stream signal-to-noise ratio (SNR) for each of the plurality of subcarriers or an average SNR for the plurality of subcarriers for each of the plurality of wireless stations. In some examples, the respective per-STA UL beamforming feedback for each of the plurality of wireless STAs is provided in single user compressed beamforming (SU-CBF) format or multi-user compressed beamforming (MU-CBF).

(68) In some examples, the wireless AP can transmit a third packet configured to cause the plurality of wireless STAs to concurrently transmit a multi-user (MU) multiple-input multiple-output (MIMO) communication in accordance with the per-STA UL beamforming feedback. For example, in the operating environment **700** of FIG. 7, the wireless AP **702** can transmit a trigger frame to cause the wireless STAs **704** to transmit beamformed UL-MU frame data in the manner illustrated in FIG. 6, described above.

(69) In some examples, the wireless AP can receive a respective number of antennas of each of the plurality of wireless STAs, prior to transmitting the first frame. For example, in the operating environment **700** of FIG. 7, the wireless AP **702** and the wireless STAs **704** may have a capability exchange when the wireless STAs **704** are associating with the wireless AP **702** in which each wireless STA **704** provides the number of antennas it may use for beamformed transmissions.

(70) In some examples, the wireless AP can transmit an indication to each of the plurality of wireless STAs, of a modification restriction for using the one or more respective unitary precoder matrices; and the indication is associated with obtaining the respective unitary precoder matrices jointly using the plurality of sounding packets from the plurality of wireless STAs. For example, in the operating environment **700** of FIG. 7, the wireless AP **702** and the wireless STAs **704** may include a flag, bit or other indication to the wireless STAs **704** to use the respective unitary precoder matrix without modification, except to the extent needed for regular beamformer operation such as phase alignment, stream weighting, etc., in a frame transmitted by the wireless AP **702**, such as frames shown in FIG. 6. This can help avoid potential interference when unitary precoder matrices for all wireless STAs **704** are jointly determined using the process shown in FIG. 5, for example.

(71) FIG. 9 shows a block diagram of an example wireless communication device **900** that supports UL MU-MIMO precoding using per-STA feedback. In some implementations, the wireless communication device **900** may be configured to perform the process **800** described above with

reference to FIG. 8. The wireless communication device **900** may be an example implementation of wireless AP **702** of FIG. 7. In some implementations, the wireless communication device **900** may be a chip, SoC, chipset, package or device that may include: one or more modems (such as, a Wi-Fi (IEEE 802.11) modem or a cellular modem such as 3GPP 4G LTE or 5G compliant modem), one or more processors, processing blocks or processing elements (collectively “the processor”); one or more radios (collectively “the radio”); and one or more memories or memory blocks (collectively “the memory”). In some implementations, the wireless communication device **900** may be a device for use in a wireless AP, such as the wireless AP **102** described above with reference to FIG. 1. In other implementations, the wireless communication device **900** may be a wireless AP that includes such a chip, SoC, chipset package or device as well as at least one antenna.

(72) In some implementations, the wireless communication device **900** may be capable of transmitting and receiving wireless communications in the form of, for example, wireless packets. For example, the wireless communication device may be configured or operable to transmit and receive packets in the form of physical layer PPDU and MPDU conforming to one or more of the IEEE 802.11 family of wireless communication protocol standards. The wireless communication device **900** may include components for bi-directional communications including components for transmitting and receiving communications, such as a communications manager **920**, an input/output (I/O) controller **910**, a transceiver **915**, an antenna **925**, a memory **930**, code **935** and a processor **940**. These components may be in electronic communication or otherwise coupled (such as operatively, communicatively, functionally, electronically, electrically) via one or more buses (such as a bus **945**).

(73) The I/O controller **910** may manage input and output signals for the wireless communication device **900**. The I/O controller **910** also may manage peripherals not integrated into the wireless communication device **900**. In some implementations, the I/O controller **910** may represent a physical connection or port to an external peripheral. In some implementations, the I/O controller **910** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. Additionally, or alternatively, the I/O controller **910** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some implementations, the I/O controller **910** may be implemented as part of a processor or processing system, such as the processor **940**. In some implementations, a user may interact with the wireless communication device **900** via the I/O controller **910** or via hardware components controlled by the I/O controller **910**.

(74) In some implementations, the wireless communication device **900** may include a single antenna **925**. However, in some other implementations, the wireless communication device **900** may have more than one antenna **925**, which may be capable of concurrently transmitting or receiving multiple wireless transmissions. The transceiver **915** may communicate bi-directionally, via the one or more antennas **925**, wired, or wireless links as described herein. For example, the transceiver **915** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **915** also may include a modem to modulate the packets, to provide the modulated packets to one or more antennas **925** for transmission, and to demodulate packets received from the one or more antennas **925**.

(75) In some implementations, the transceiver **915** may include one or more interfaces, such as one or more interfaces coupled with the one or more antennas **925** that are configured to support various receiving or obtaining operations, or one or more interfaces coupled with the one or more antennas **925** that are configured to support various transmitting or outputting operations, or a combination thereof. In some implementations, the transceiver **915** may include or be configured for coupling with one or more processors or memory components that are operable to perform or support operations associated with received or obtained information or signals, or to generate information or other signals for transmission or other outputting, or any combination thereof. In some implementations, the transceiver **915**, or the transceiver **915** and the one or more antennas

925, or the transceiver **915** and the one or more antennas **925** and one or more processors or memory components (for example, the processor **940**, or the memory **930**, or both), may be included in a chip or chip assembly that is installed in the wireless communication device **900**.

(76) The memory **930** may include random access memory (RAM) and read-only memory (ROM). The memory **930** may store computer-readable, computer-executable code **935** including instructions that, when executed by the processor **940**, cause the wireless communication device **900** to perform various functions described herein. The code **935** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some implementations, the code **935** may not be directly executable by the processor **940** but may cause a computer (for example, when compiled and executed) to perform functions described herein. In some implementations, the memory **930** may contain, among other things, a basic I/O system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices.

(77) The processor **940** may be any one or more suitable processors capable of executing scripts or instructions of one or more software programs stored in the wireless communication device **900** (such as within the memory **930**). In some implementations, the processor **940** may be a component of a processing system. A processing system may generally refer to a system or series of machines or components that receives inputs and processes the inputs to produce a set of outputs (which may be passed to other systems or components of, for example, the wireless communication device **900**). For example, a processing system of the wireless communication device **900** may refer to a system including the various other components or subcomponents of the wireless communication device **900**, such as the processor **940**, or the transceiver **915**, or the communications manager **920**, or other components or combinations of components of the wireless communication device **900**.

(78) The processing system of the wireless communication device **900** may interface with other components of the wireless communication device **900**, and may process information received from other components (such as inputs or signals) or output information to other components. For example, a chip or modem of the wireless communication device **900** may include a processing system, a first interface to output information and a second interface to obtain information. In some implementations, the first interface may refer to an interface between the processing system of the chip or modem and a transmitter, such that the wireless communication device **900** may transmit information output from the chip or modem. In some implementations, the second interface may refer to an interface between the processing system of the chip or modem and a receiver, such that the wireless communication device **900** may obtain information or signal inputs, and the information may be passed to the processing system. A person having ordinary skill in the art will readily recognize that the first interface also may obtain information or signal inputs, and the second interface also may output information or signal outputs.

(79) The communications manager **920** may support wireless communication by wireless communication device **900** in accordance with examples as disclosed herein. In some implementations, the communications manager **920** may be configured to perform various operations (for example, receiving, monitoring, transmitting) using or otherwise in cooperation with the transceiver **915**, the one or more antennas **925**, or any combination thereof. Although the communications manager **920** is illustrated as a separate component, in some implementations, one or more functions described with reference to the communications manager **920** may be supported by or performed by the processor **940**, the memory **930**, the code **935**, or any combination thereof. For example, the code **935** may include instructions executable by the processor **940** to cause the wireless communication device **900** to perform various aspects of conflict mitigation for EMLSR links between multi-link devices as described herein, or the processor **940** and the memory **930** may be otherwise configured to perform or support such operations.

(80) In some examples, the communications manager **920** may be configured as or otherwise support a means for transmitting a first frame on a wireless channel, the first frame identifying a

plurality of wireless stations (STAs) that are to perform uplink (UL) sounding of the wireless channel.

(81) In some examples, the communications manager **920** may be configured as or otherwise support a means for receiving via the wireless channel, associated with the transmission of the first frame, multiple sounding packets, the multiple sounding packets including a respective sounding packet from each of multiple wireless stations among the plurality of wireless stations.

(82) In some examples, the communications manager **920** may be configured as or otherwise support a means for transmitting a second frame on the wireless channel associated with the receipt of the multiple sounding packets, the second frame including per-STA UL beamforming feedback for a wireless station among the multiple wireless stations.

(83) Implementation examples are described in the following numbered clauses:

(84) Clause 1. A wireless access point (AP), comprising: a processing system that includes processor circuitry and memory circuitry that stores code, the processing system configured to cause the wireless AP to: transmit a first packet that indicates to each of a plurality of wireless stations (STAs) to transmit an uplink (UL) sounding packet; receive, in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively; and transmit a second packet that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA.

(85) Clause 2. The wireless AP of clause 1, wherein the processing system is further configured to cause the wireless AP to: perform the measurements of the plurality of sounding packets over a plurality of subcarriers; obtain a respective channel matrix for each of the plurality of subcarriers in accordance with the measurements; obtain a respective projection of each channel matrix for each of the plurality of wireless STAs in accordance with a respective linear equalizer; and obtain the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs in accordance with respective block diagonal components of the respective projections.

(86) Clause 3. The wireless AP of clause 2, wherein the processing system is further configured to cause the wireless AP to compress the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs using a respective Givens rotation operation to generate respective angles, wherein the per-STA UL beamforming feedback includes the respective angles.

(87) Clause 4. The wireless AP of any one of clauses 2-3, wherein, to obtain the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs, the processing system is configured to cause the wireless AP to perform a singular value decomposition (SVD), Eigen Value Decomposition (EVD) or QR decomposition operation on the respective block diagonal components of the respective projections.

(88) Clause 5. The wireless AP of any one of clauses 2-4, wherein, the respective projection of each channel matrix for each of the plurality of wireless STAs comprises a null space matrix projection.

(89) Clause 6. The wireless AP of clause 5, wherein the null space matrix projection comprises a null space block diagonal matrix projection.

(90) Clause 7. The wireless AP any one of clauses 1-6, wherein the first packet includes a trigger configured to cause the plurality of wireless STAs to transmit the plurality of sounding packets.

(91) Clause 8. The wireless AP of any one of clauses 1-7, wherein the second packet includes a respective compressed beamforming report field for each of the plurality of wireless STAs that includes the per-STA UL beamforming feedback for the plurality of wireless STAs.

(92) Clause 9. The wireless AP of any one of clauses 1-8, wherein the respective per-STA UL beamforming feedback for each of the plurality of wireless STAs is provided in single user compressed beamforming (SU-CBF) format or multi-user compressed beamforming (MU-CBF).

(93) Clause 10. The wireless AP of any one of clauses 1-9, wherein the processing system is further configured to cause the wireless AP to transmit a third packet configured to cause the plurality of

wireless STAs to concurrently transmit a multi-user (MU) multiple-input multiple-output (MIMO) communication in accordance with the per-STA UL beamforming feedback.

(94) Clause 11. The wireless AP of any one of clauses 1-10, wherein the second packet further includes a respective per stream signal-to-noise ratio (SNR) for each of the plurality of subcarriers or an average SNR for the plurality of subcarriers for each of the plurality of wireless STAs.

(95) Clause 12. The wireless AP of any one of clauses 1-11, wherein the processing system is configured to cause the wireless AP to receive a respective number of antennas of each of the plurality of wireless STAs, prior to transmitting the first frame.

(96) Clause 13. The wireless AP of any one of clauses 1-12, wherein the processing system is configured to cause the wireless AP to transmit an indication to each of the plurality of wireless STAs, of a modification restriction for using the one or more respective unitary precoder matrices.

(97) Clause 14. A method for wireless communication by a wireless access point (AP), comprising: transmitting a first packet that indicates to each of a plurality of wireless stations (STAs) to transmit an uplink (UL) sounding packet; receiving, in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively; and transmitting a second frame that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA.

(98) Clause 15. The method of clause 14, further comprising: performing the measurements of the plurality of sounding packets over a plurality of subcarriers; obtaining a respective channel matrix for each of the plurality of subcarriers in accordance with the measurements; obtaining a respective projection of each channel matrix for each of the plurality of wireless STAs in accordance with a respective linear equalizer; and obtaining the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs in accordance with respective block diagonal components of the respective projections.

(99) Clause 16. The method of clause 15, wherein further comprising compressing the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs using a respective Givens rotation operation to generate respective angles, wherein the per-STA UL beamforming feedback includes the respective angles.

(100) Clause 17. The method of any one of clauses 15-16, wherein obtaining the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs comprises performing a singular value decomposition (SVD), Eigen Value Decomposition (EVD) or QR decomposition operation on the respective block diagonal components of the respective projections.

(101) Clause 18. The method of any one of clauses 15-17, wherein the respective projection of each channel matrix for each of the plurality of wireless STAs comprises a null space matrix projection.

(102) Clause 19. The method of clause 18, wherein the null space matrix projection comprises a null space block diagonal matrix projection.

(103) Clause 20. The method of any one of clauses 14-19, wherein the first packet includes a trigger configured to cause the plurality of wireless STAs to transmit the plurality of sounding packets.

(104) Clause 21. The method of any one of clauses 14-20, wherein the second packet includes a respective compressed beamforming report field for each of the plurality of wireless STAs that includes the per-STA UL beamforming feedback for the plurality of wireless STAs.

(105) Clause 22. The method of any one of clauses 14-21, wherein the respective per-STA UL beamforming feedback for each of the plurality of wireless STAs is provided in single user compressed beamforming (SU-CBF) format or multi-user compressed beamforming (MU-CBF).

(106) Clause 23. The method of any one of clauses 14-23, comprising transmitting a third packet configured to cause the plurality of wireless STAs to concurrently transmit a multi-user (MU)

multiple-input multiple-output (MIMO) communication in accordance with the per-STA UL beamforming feedback.

(107) Clause 24. The method of any one of clauses 14-23, wherein the second packet further includes a respective per stream signal-to-noise ratio (SNR) for each of the plurality of subcarriers or an average SNR for the plurality of subcarriers for each of the plurality of wireless STAs.

(108) Clause 25. The method of any one of clauses 14-24, further comprising receiving a respective number of antennas of each of the plurality of wireless STAs, prior to transmitting the first frame.

(109) Clause 26. The method of any one of clauses 14-25, further comprising transmitting an indication to each of the plurality of wireless STAs, of a modification restriction for using the one or more respective unitary precoder matrices.

(110) Clause 27. An apparatus for a wireless access point (AP), comprising: means for transmitting a first packet that indicates to each of a plurality of wireless stations (STAs) to transmit an uplink (UL) sounding packet; means for receiving, in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively; and means for transmitting a second frame that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA.

(111) Clause 28. The apparatus of clause 27, further comprising: means for performing the measurements of the plurality of sounding packets over a plurality of subcarriers; means for obtaining a respective channel matrix for each of the plurality of subcarriers in accordance with the measurements; means for obtaining a respective projection of each channel matrix for each of the plurality of wireless STAs in accordance with a respective linear equalizer; and means for obtaining the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs in accordance with respective block diagonal components of the respective projections.

(112) Clause 29. The apparatus of clause 28, further comprising means for compressing the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs using a respective Givens rotation operation to generate respective angles, wherein the per-STA UL beamforming feedback includes the respective angles.

(113) Clause 30. A non-transitory computer-readable medium storing instructions for communication by a wireless access point (AP), the instructions including code to: transmit a first packet that indicates to each of a plurality of wireless stations (STAs) to transmit an uplink (UL) sounding packet; receive, in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively; and transmit a second packet that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA.

(114) Clause 31. A wireless access point (AP), comprising: a processing system that includes processor circuitry and memory circuitry that stores code, the processing system configured to cause the wireless AP to: receive a plurality of sounding packets from a plurality of wireless stations (STAs); perform measurements of the plurality of sounding packets; for each of a plurality of subcarriers: determine a respective channel matrix associated with a subcarrier, the respective channel matrix representative of an estimated channel for the plurality of wireless STAs for the subcarrier and based on the measurements of the plurality of sounding packets; and obtain a respective equalizer matrix based on respective channel matrix associated with the subcarrier; determine a respective modified channel matrix associated with the subcarrier, the channel matrix comprising a projection of the respective channel matrix on the respective equalizer matrix; and determining a respective unitary precoder matrix for each of the plurality of wireless STAs based

on diagonal components of the modified channel matrix respectively associated with the plurality of wireless STAs; and transmitting UL beamforming feedback for each of the plurality of wireless STAs, the UL beamforming feedback representative of the respective unitary precoder matrix for each of the plurality of wireless STAs.

(115) As used herein, the term “determine” or “determining” encompasses a wide variety of actions and, therefore, “determining” can include calculating, computing, processing, deriving, estimating, investigating, looking up (such as via looking up in a table, a database, or another data structure), inferring, ascertaining, or measuring, among other possibilities. Also, “determining” can include receiving (such as receiving information), accessing (such as accessing data stored in memory) or transmitting (such as transmitting information), among other possibilities. Additionally, “determining” can include resolving, selecting, obtaining, choosing, establishing and other such similar actions.

(116) As used herein, a phrase referring to “at least one of” or “one or more of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover: a, b, c, a-b, a-c, b-c, and a-b-c. As used herein, “or” is intended to be interpreted in the inclusive sense, unless otherwise explicitly indicated. For example, “a or b” may include a only, b only, or a combination of a and b. Furthermore, as used herein, a phrase referring to “a” or “an” element refers to one or more of such elements acting individually or collectively to perform the recited function(s). Additionally, a “set” refers to one or more items, and a “subset” refers to less than a whole set, but non-empty.

(117) As used herein, “based on” is intended to be interpreted in the inclusive sense, unless otherwise explicitly indicated. For example, “based on” may be used interchangeably with “based at least in part on,” “associated with,” “in association with,” or “in accordance with” unless otherwise explicitly indicated. Specifically, unless a phrase refers to “based on only ‘a,’” or the equivalent in context, whatever it is that is “based on ‘a,’” or “based at least in part on ‘a,’” may be based on “a” alone or based on a combination of “a” and one or more other factors, conditions, or information.

(118) The various illustrative components, logic, logical blocks, modules, circuits, operations, and algorithm processes described in connection with the examples disclosed herein may be implemented as electronic hardware, firmware, software, or combinations of hardware, firmware, or software, including the structures disclosed in this specification and the structural equivalents thereof. The interchangeability of hardware, firmware and software has been described generally, in terms of functionality, and illustrated in the various illustrative components, blocks, modules, circuits and processes described above. Whether such functionality is implemented in hardware, firmware or software depends upon the particular application and design constraints imposed on the overall system.

(119) Various modifications to the examples described in this disclosure may be readily apparent to persons having ordinary skill in the art, and the generic principles defined herein may be applied to other examples without departing from the spirit or scope of this disclosure. Thus, the claims are not intended to be limited to the examples shown herein, but are to be accorded the widest scope consistent with this disclosure, the principles and the novel features disclosed herein.

(120) Additionally, various features that are described in this specification in the context of separate examples also can be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation also can be implemented in multiple examples separately or in any suitable subcombination. As such, although features may be described above as acting in particular combinations, and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

(121) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in

sequential order, or that all illustrated operations be performed, to achieve desirable results. Further, the drawings may schematically depict one or more example processes in the form of a flowchart or flow diagram. However, other operations that are not depicted can be incorporated in the example processes that are schematically illustrated. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the illustrated operations. In some circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the examples described above should not be understood as requiring such separation in all examples, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

Claims

1. A wireless access point (AP), comprising: a processing system that includes processor circuitry and memory circuitry that stores code, the processing system configured to cause the wireless AP to: transmit a first packet that indicates to each of a plurality of wireless stations (STAs) to transmit an uplink (UL) sounding packet; receive, in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively; and transmit a second packet that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA.
2. The wireless AP of claim 1, wherein the processing system is further configured to cause the wireless AP to: perform the measurements of the plurality of sounding packets over a plurality of subcarriers; obtain a respective channel matrix for each of the plurality of subcarriers in accordance with the measurements; obtain a respective projection of each channel matrix for each of the plurality of wireless STAs in accordance with a respective linear equalizer; and obtain the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs in accordance with respective block diagonal components of the respective projections.
3. The wireless AP of claim 2, wherein the processing system is further configured to cause the wireless AP to compress the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs using a respective Givens rotation operation to generate respective angles, wherein the per-STA UL beamforming feedback includes the respective angles.
4. The wireless AP of claim 2, wherein, to obtain the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs, the processing system is configured to cause the wireless AP to perform a singular value decomposition (SVD), Eigen Value Decomposition (EVD) or QR decomposition operation on the respective block diagonal components of the respective projections.
5. The wireless AP of claim 2, wherein, the respective projection of each channel matrix for each of the plurality of wireless STAs comprises a null space matrix projection.
6. The wireless AP of claim 5, wherein the null space matrix projection comprises a null space block diagonal matrix projection.
7. The wireless AP of claim 1, wherein the first packet includes a trigger configured to cause the plurality of wireless STAs to transmit the plurality of sounding packets.
8. The wireless AP of claim 1, wherein the second packet includes a respective compressed beamforming report field for each of the plurality of wireless STAs that includes the per-STA UL beamforming feedback for the plurality of wireless STAs.
9. The wireless AP of claim 1, wherein the respective per-STA UL beamforming feedback for each of the plurality of wireless STAs is provided in single user compressed beamforming (SU-CBF) format or multi-user compressed beamforming (MU-CBF).

10. The wireless AP of claim 1, wherein the processing system is further configured to cause the wireless AP to transmit a third packet configured to cause the plurality of wireless STAs to concurrently transmit a multi-user (MU) multiple-input multiple-output (MIMO) communication in accordance with the per-STA UL beamforming feedback.
11. The wireless AP of claim 1, wherein the second packet further includes a respective per stream signal-to-noise ratio (SNR) for each of the plurality of subcarriers or an average SNR for the plurality of subcarriers for each of the plurality of wireless STAs.
12. The wireless AP of claim 1, wherein the processing system is configured to cause the wireless AP to receive a respective number of antennas of each of the plurality of wireless STAs, prior to transmitting the first packet.
13. The wireless AP of claim 1, wherein the processing system is configured to cause the wireless AP to transmit an indication to each of the plurality of wireless STAs, of a modification restriction for using the one or more respective unitary precoder matrices; and the indication is associated with the processing system obtaining the respective unitary precoder matrices jointly using the plurality of sounding packets from the plurality of wireless STAs.
14. A method for wireless communication by a wireless access point (AP), comprising: transmitting a first packet that indicates to each of a plurality of wireless stations (STAs) to transmit an uplink (UL) sounding packet; receiving, in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively; and transmitting a second packet that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA.
15. The method of claim 14, further comprising: performing the measurements of the plurality of sounding packets over a plurality of subcarriers; obtaining a respective channel matrix for each of the plurality of subcarriers in accordance with the measurements; obtaining a respective projection of each channel matrix for each of the plurality of wireless STAs in accordance with a respective linear equalizer; and obtaining the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs in accordance with respective block diagonal components of the respective projections.
16. The method of claim 15, wherein further comprising compressing the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs using a respective Givens rotation operation to generate respective angles, wherein the per-STA UL beamforming feedback includes the respective angles.
17. The method of claim 15, wherein obtaining the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs comprises performing a singular value decomposition (SVD), Eigen Value Decomposition (EVD) or QR decomposition operation on the respective block diagonal components of the respective projections.
18. The method of claim 15, wherein the respective projection of each channel matrix for each of the plurality of wireless STAs comprises a null space matrix projection.
19. The method of claim 18, wherein the null space matrix projection comprises a null space block diagonal matrix projection.
20. The method of claim 14, wherein the first packet includes a trigger configured to cause the plurality of wireless STAs to transmit the plurality of sounding packets.
21. The method of claim 14, wherein the second packet includes a respective compressed beamforming report field for each of the plurality of wireless STAs that includes the per-STA UL beamforming feedback for the plurality of wireless STAs.
22. The method of claim 14, wherein the respective per-STA UL beamforming feedback for each of the plurality of wireless STAs is provided in single user compressed beamforming (SU-CBF) format or multi-user compressed beamforming (MU-CBF).

23. The method of claim 14, comprising transmitting a third packet configured to cause the plurality of wireless STAs to concurrently transmit a multi-user (MU) multiple-input multiple-output (MIMO) communication in accordance with the per-STA UL beamforming feedback.
24. The method of claim 14, wherein the second packet further includes a respective per stream signal-to-noise ratio (SNR) for each of the plurality of subcarriers or an average SNR for the plurality of subcarriers for each of the plurality of wireless STAs.
25. The method of claim 14, further comprising receiving a respective number of antennas of each of the plurality of wireless STAs, prior to transmitting the first packet.
26. The method of claim 14, further comprising transmitting an indication to each of the plurality of wireless STAs, of a modification restriction for using the one or more respective unitary precoder matrices; and the indication is associated with obtaining the respective unitary precoder matrices jointly using the plurality of sounding packets from the plurality of wireless STAs.
27. An apparatus for a wireless access point (AP), comprising: means for transmitting a first packet that indicates to each of a plurality of wireless stations (STAs) to transmit an uplink (UL) sounding packet; means for receiving, in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively; and means for transmitting a second packet frame that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA.
28. The apparatus of claim 27, further comprising: means for performing the measurements of the plurality of sounding packets over a plurality of subcarriers; means for obtaining a respective channel matrix for each of the plurality of subcarriers in accordance with the measurements; means for obtaining a respective projection of each channel matrix for each of the plurality of wireless STAs in accordance with a respective linear equalizer; and means for obtaining the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs in accordance with respective block diagonal components of the respective projections.
29. The apparatus of claim 28, further comprising means for compressing the one or more respective unitary precoder matrices associated with each of the plurality of wireless STAs using a respective Givens rotation operation to generate respective angles, wherein the per-STA UL beamforming feedback includes the respective angles.
30. A non-transitory computer-readable medium storing instructions for communication by a wireless access point (AP), the instructions including code to: transmit a first packet that indicates to each of a plurality of wireless stations (STAs) to transmit an uplink (UL) sounding packet; receive, in association with the transmission of the first packet, a plurality of sounding packets from the plurality of wireless STAs, respectively; and transmit a second packet that includes respective per-STA UL beamforming feedback for each of the plurality of wireless STAs in accordance with measurements of the received sounding packets, the respective per-STA UL beamforming feedback being representative of one or more respective unitary precoder matrices associated with the respective wireless STA.
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